

1. Unite/Prime V2 Regression Suite – Overview

 Page is about?

Purpose

The V2 nightly regression suite validates core 529 and ABLE plan functionality across enrollment, contribution, withdrawal, account maintenance, balance validation, web registration, web login, and Sardine (fraud-detection enrollment) flows. It runs every weeknight against Stage 1 to catch regressions introduced by daily code deployments before they reach downstream environments.

Why We Run It

- **Release gate:** V2 regression is the primary automated validation gate for the monolith application on Stage 1. Failures here trigger bug reporting and main-branch lock per the Bug Handling process.
- **Coverage breadth:** Covers ~95% of core 529 flows across 10+ plans (NYD, NYA, NYB, MID, MND, AKB, ILB, CAD, TNB, PAB, and more).
- **Stability signal:** Nightly pass/fail trend is the leading indicator for release readiness and environment health.

Prime V2 – Nightly Regression Suite Overview

Owner	QA Automation Team (AM Squad)
Environment	Stage 1
Version	Prime V2 (Java + Selenium + TestNG + Cucumber, Ant build)
Execution	Nightly (Mon–Fri) via Jenkins
Jenkins Job	STAGE1-Daily-Unite-Prime-Regression
Schedule	Daily @ 12:00 AM – 1:00 AM EST (Mon–Fri).
TestNG Reports	View Latest HTML Reports
qTest Dashboard	Regression Execution Dashboard
Process Reference	Automation Naming + Tagging Standards
Jira	Failures logged to JIRA and reviewed under CI lock policy before 10 AM EST. Branch lock and revert governed by Fail-Safe Protocol for stability.

High-Level Structure

The V2 regression is organized into **8 active daily modules**, each with its own TestNG suite XML:

Category /Suites	Suite Count	Purpose
Daily (Active)	9	Nightly front-office regression against Stage 1, core modules automated across 529, Advisor, ABLE, and CSR flows
Smoke (Stage 5 / Stage 2 / QC4)	9	Quick-validation for CAT / partner environments
Release (Stage 1 / Stage 2)	3	Release-gate regression (<i>TODO</i>)
Weekly (Backoffice)	2	Tuesday & Wednesday backoffice feed processing (<i>TODO</i>)
Other (Specialty)	3	Flywire, SSgA Conversion, ACH Debit (<i>TODO</i>)

Active Daily Modules (9)

#	Module	Suite XML	Tests	Approx Duration	Thread Count	Note
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1	Enrollment	stage1-enrollments.xml	28 tests / 119 methods	~3 hrs	3	
2	Legacy Web Registration	stage1-web-registration.xml	11 tests / 54 methods	~10 min	1	
3	Legacy Web Login	stage1-web-login.xml	19 tests / 27 methods	~15 min	1	
4	CSR Account Maintenance	stage1-csr-acct-maintenance.xml	14 tests / 24 methods	~3.5 hrs	1	
5	Contributions	stage1-contributions.xml	15 tests / 49 methods	~9 hrs	4	
6	Withdrawals	stage1-withdrawals.xml	14 tests / 73 methods	~4.5–6 hrs	4	
7	Account Balance Page	stage1-acct-overview.xml	10 tests / 10 methods	—	4	
8	Investment Options	stage1-investment-options.xml	2 tests	—	1	Added a baseline not covered fully
9	Transfer	stage1-transfer.xml	2 tests	—	1	Added a baseline not covered fully
10	Sardine Regression	stage1-sardine-regression.xml	10 tests / 29 methods	~3.5 hrs	1	Not owned by AM Squad team, added to nightly regression as well
11	Empower Plan	stage1-empower-plan.xml	20 tests	—	4	Not owned by AM Squad team, added to nightly regression as well

Total active daily: ~143 tests / ~454+ methods (scenarios) executed nightly.

Approximate Total Execution Time

Based on available duration data (from Confluence PDFs and sequential/parallel execution):

Module	Duration
Balance Page	~45 Mins
Contributions	~3 Hrs
CSR Account Maintenance	~1 Hrs
Enrollment	~7 hrs
Legacy Web Registration	~55 min
Legacy Web Login	~55 min
Withdrawals	~4 hrs
Sardine	~3.5 hrs
Investment Options	~15 Mins
Empower	~4 Hrs

Total estimated nightly wall-clock time depends on parallelization across Jenkins executors. Individual module durations suggest **~6-6.5 hours of sequential test time**, compressed by parallel execution across modules and within modules (thread counts 1–4).

Best Practices & Governance

Area	Practice
Plans Covered	NYD, NYA, NYB, MID, MND, ILB, MIA, AKB, CAD, TNB, PAB (extended)
Folder Structure	Unified under project-specific runners and plan folders.
Tagging Convention	@regression, @smoke, @functional, @dailyrun.
Coverage Tracking	Synced to qTest via Regression Dashboard.
Branch Strategy	CI/CD merge requests only after successful daily regression.
Reporting	Auto export of TestNG HTML reports and qTest sync.

Flakiness Policy	Failed tests triaged before next run; unstable cases quarantined.
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Key Highlights

- **qTest and TestNG integration** for traceability and reporting.
- **Flakiness Tracking** and Fail-Safe protocol maintained for CI stability.
- **Module-per-XML**: Each module has its own TestNG suite XML under Regression Suites XMLs/daily/. This replaced the older monolithic stage1-frontoffice.xml (now in Archive/).
- **Runner classes**: `com.cs529.qa.prime.runner.FeatureRunner` (sequential) and `com.cs529.qa.prime.runner.ParallelFeatureRunner` (parallel). Most modules use one or both.
- **Tags**: Tests are filtered by `@regression` and `@dailyrun` (plus module-specific tags like `@sardinerun`, `@ablerun`, `@dailyrun-naa`, `@dailyrun-non-naa`, `@agebasedfundrun`).
- **Parameters**: Each test block specifies `traunch` (plan code, e.g. `nyd`, `mid`, `akb`) and `features` (path to `.feature` file).
- **HTML reports**: Available in `10_IMPORTS_RAW/AM_Regression_Reports/asRegressionTestSuite - <Module> {1,2,3}.html`. The three numbered files per module are different views of the same execution run.
- **Newly added modules** (not in old Confluence docs): **Investment Options** (`stage1-investment-options.xml`), **Transfer** (`stage1-trasfer.xml`) and **Empower Plan** (`stage1-empower-plan.xml`).

Next Steps / Planned Expansions

- Add **release-specific smoke, plan conversion, API Test Suites and performance suites**.
- Build **API, Mobile, Performance testing, V3 unite/Entity and SFRP baseline pipelines** for 2026.
- Continue standardizing test data provisioning and auto-retry framework.